

Data Analysis and Surrogate Modeling of a Carbon Capture Process

1. Project Background

Reducing carbon emissions is necessary for a sustainable future. In this context, carbon capture technology, which allows for the capture of CO₂ from the atmosphere and helps mitigate climate change, has received significant attention in recent years. By varying the input variables, such as temperature, humidity, adsorption time, desorption time, and steam velocity, the resulting CO₂ adsorption/desorption quantities change. Currently, a research lab at Georgia Tech is generating a large carbon capture dataset by modifying those input variables. In this project, we aim to investigate different surrogate modeling techniques and identify which of them are the most appropriate to predict the adsorption and desorption CO₂ quantities based on the mentioned input variables.

1.1. Dataset description

With the approval of data sourcing from a Georgia Institute of Technology research laboratory, the dataset models carbon dioxide captures on adiabatic shell fibers. The five input and two output variables are all continuous and are presented in Table 1. Additionally, our dataset contains 20,673 data points and is a dynamic dataset, due to the ongoing research at Georgia Tech (Lee et al., 2023).

Inputs	Description	Units	Range of inputs
T	Temperature	K	225 – 299.15
Humidity	Humidity	%	3.6 – 80
AdsTime	Adsorption Time	s	1495 – 6000
DesTime	Desorption Time	s	301 – 4994
SteamVelocity	Steam Velocity	m/s	1E-2 – 1E-1
Outputs	Description	Units	Range of Outputs
q_ads	Adsorption Quantity	mol/kg	4E-1 – 1.4
q_des	Desorption Quantity	mol/kg	6E-4 – 1.2

Table 1. Carbon capture dataset variables and their respective ranges.

For data preparation, we handled missing and null values by removing rows using the `dropna()` method and applied winsorization to cap extreme outliers to reduce their impact.

2. Project Goal

In previous work, we performed surface approximation for the carbon capture process, which was described in Section 1.1. Various surrogate modeling techniques were compared in terms of their performance metrics and their ability to be embedded in optimization frameworks. Those surrogate

models were compared only in terms of their R-squared and root mean squared error (RMSE) metrics, and the metrics and the parity plots were not adequate for most of them. This leaves open the question of whether the given dataset requires pre-processing and whether alternative techniques can be used for the surface approximation of the carbon capture process.

In this work, we are building surrogate models that approximate the two outputs of the carbon capture dataset that balance predictive ability and implementation efficiency. Data management techniques will be utilized to assess whether the results of the previous work are reliable, and to improve the performance of the new models that will be constructed. Additionally, other surrogate modeling techniques will be investigated, and their hyperparameters will be optimized. The impact of different train/test splits on the accuracy of the models will also be assessed in the context of this work. Since the dataset contains a large number of points, we will investigate whether a smaller number of instances can be chosen to build more complex surrogate models that are reliable. The goal of this work is to find appropriate surrogate models that strike a balance between predictive accuracy (R^2) and computational efficiency (training time), and for this reason, we define an aggregate score that considers both these two metrics.

2.1. Project Strategy

2.1.1 Dataset Management - Winsorization

Winsorization is a statistical technique used to limit the impact of outliers by capping extreme values within a dataset at defined percentiles, improving the accuracy of analyses without excluding data. Before applying it, we identified variables with large variability, such as desorption time, DesTime [s] (input variable) and desorption rate, q_{des} [mol/kg] (output variable), through boxplots. Using a 2% threshold, we significantly reduced the variability in both variables as shown below.

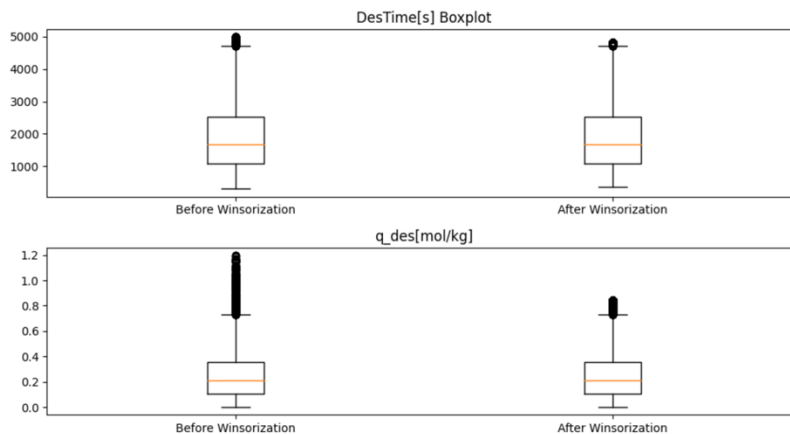


Figure 1. Boxplot of DesTime [s] and q_{des} [mol/kg] before and after Winsorization.

A preliminary analysis compared the model performance of a LASSO model utilizing winsorized and unprocessed testing dataset from the 80% training dataset split, and in both cases the model was validated with the same 20% unprocessed validation set. This step is included as a part of baseline model workflow. Winsorization slightly improved R^2 (+0.015) for desorption rate, but

worsened MSE, RMSE, and Maximum Absolute Error. Despite these irregularities, parity plots for desorption quantity on the validation set improved significantly (less datapoints deviated from the parity line), as shown in Table 2. Note the parity for quantity of adsorption does not show any difference as the variables being winsorized (DesTime [s] and q_{des} [mol/kg]) does not highly impact q_{ads} prediction which later is explained using our results from LASSO, ALAMO as well as correlation matrix.

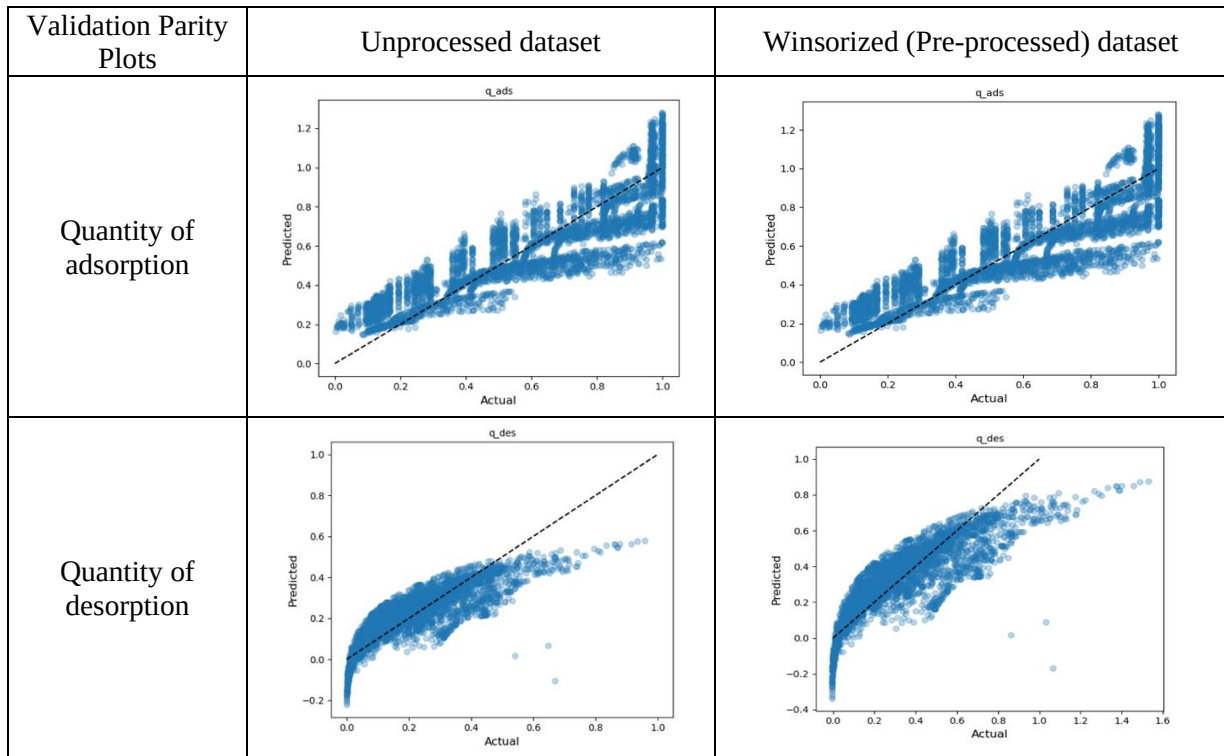


Table 2. Parity plots on the validation set of desorption rate prediction comparison between Winsorized and unprocessed data.

2.2. Surrogate modeling techniques

We will use a variety of surrogate modeling techniques to approximate the outputs of the carbon capture process, such as L1 (LASSO) regularization, ALAMO (Automated Learning of Algebraic Models), Kriging, Linear Regression using RBF transformation, and Artificial Neural Networks (ANNs) techniques. Our goal is to identify the surrogate models that strike a balance between predictive ability and computational efficiency. Since the dataset consists of more than 20,000 data points, the size of the training sets should be carefully chosen, to build complex surrogate models that require large times for their training.

3. Baseline Model and Workflow

3.1 Workflow

The workflow in Figure 2 depicts the general procedure of pre-processing, training, and then testing the raw data via the LASSO regression technique. Initially, null datapoints are removed to ensure reliable model creation. Then, the dataset is split into two subsets: 80% is allocated for training, while the remaining 20% is designated for validation. The training set is winsorized to mitigate the effect of extreme outliers on the construction of the model. Following, the processed data is used to train a LASSO model, which is particularly effective in preventing overfitting. The model is then tested with the unwinsorized validation set, with R^2 and RMSE values being reported as quality metrics. To visualize the performance, parity plots are also generated where having less datapoint deviation from the parity line indicates a better parity plot and vice versa. To investigate the effectiveness of winsorization, an analogous procedure is performed for the unwinsorized training set as well. (This was discussed in Section 2.1.1 and Table 2.)

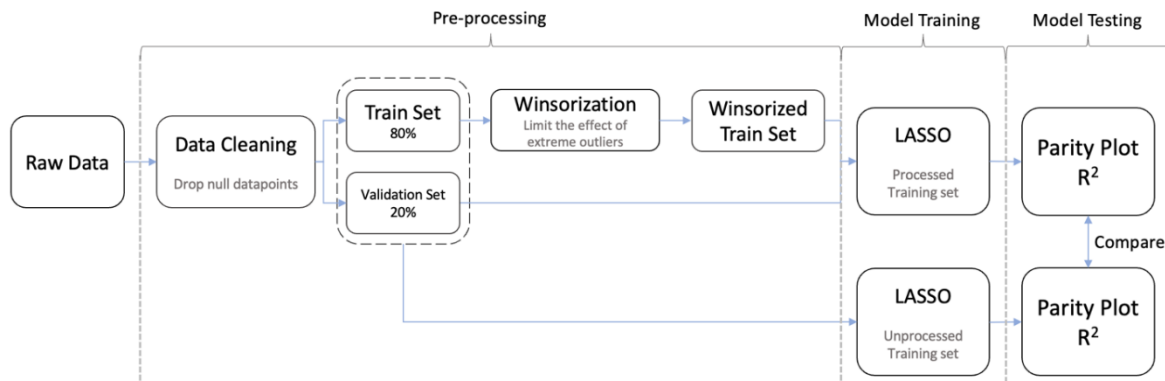


Figure 2. Baseline model workflow

3.2 LASSO Baseline Model Performance

LASSO regression is selected to train our baseline model, since L1 regularization keeps only the important input variables in the model, and a simple linear model can be constructed. In future work, the models that will be constructed using various surrogate modelling techniques will be compared to this model. Our aim is to build surrogates that will have better predictive ability and computational efficiency than the baseline model. In Figure 3, the parity plots on the validation set for the LASSO baseline models of the two outputs are presented. The R^2 values of the models on the test set are 0.775 and 0.711 for the quantity of adsorption and quantity of desorption outputs respectively, while the corresponding RMSE values stand at 0.14 and 0.13.

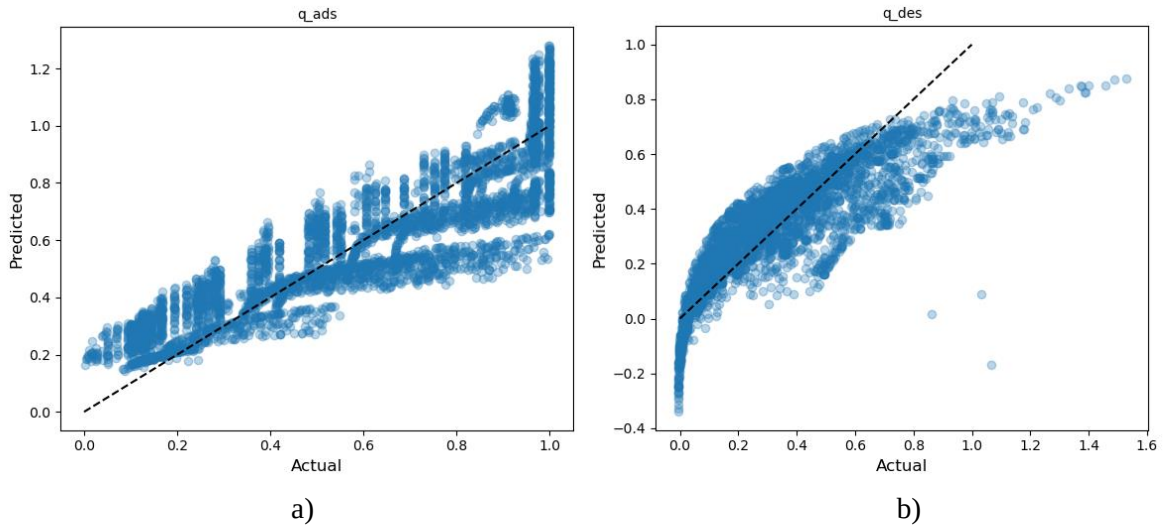


Figure 3. Parity plots on the validation set for the LASSO baseline models of a) the quantity of adsorption output and b) the quantity of desorption output.

In this work, the optimal value of LASSO's regularization coefficient α is found using Scikit-learn's 'LassoLarsIC' method. During this process, a Least Angle Regression problem is solved to minimize the Bayesian Information Criterion. Notably, for the baseline model of the quantity of adsorption, the α takes a value of zero, which means that no regularization is introduced in this model. The expression for this output is given below:

$$q_{ads} = 0.38 - 0.07 * T - 0.01 * Humidity + 0.94 * AdsTime - 0.19 * DesTime - 0.17 * SteamVelocity$$

As for baseline model of the quantity of desorption, the α stands at 0.001, which means that L1 regularization is introduced. In this case the coefficients for the temperature and humidity inputs are zero. The expression of the surrogate for the quantity of desorption is given below:

$$q_{des} = 0.81 + 0.13 * AdsTime - 0.73 * DesTime - 0.51 * SteamVelocity$$

The parity plots and the R^2 values for the two outputs indicate that more complex surrogate models with improved predictive accuracy can be built.

4. Improved model

In this study, we compared various surrogate models to evaluate their effectiveness in approximating a CO2 sorption process, focusing on balancing accuracy and training time. To aid in our decision-making, we introduced a new metric called the aggregate score, which takes both factors into account and is defined as follows:

$$\text{Aggregate Score} = 2 * (R^2 \text{ score}) - 0.001 * (\text{Training time [s]})$$

Model	Training Dataset Size (16538 total)	Training Time [s]		R ² score on validation set		Aggregate Score	
		Ads	Des	Ads	Des	Ads	Des
LASSO (Baseline)	16538	0.1	0.04	0.775	0.711	1.550	1.422
Kriging	2000	959	1559	0.518	0.153	-0.599	-0.206
ALAMO	16538	18.0	18.0	0.883	0.975	1.748	1.932
ANN	16538	144	131	0.886	0.981	1.628	1.831
RBF	16538	34.7	34.3	0.954	0.964	1.873	1.894

Table 3. Performance metrics of baseline and surrogate models

Note: All models used the same Winsorized training dataset (80%) and unprocessed validation set (20%).

4.1 Kriging

Since kriging is intended for use in geospatial statistics, its employment for our use case is atypical and some adjustments had to be made to the method, such as a reduced training set due to long computation times. To utilize the scikit-learn python package, the model used for kriging was a combination of two simpler models that allowed us to functionally mimic a kriging model. For a typical kriging model, the covariance function takes the form of:

$$k(x, x') = \sigma^2 R(x, x')$$

where sigma is the process variance and $R(x, x')$ is the correlation function. The model employed for this project is a combination of a constant kernel, which represents sigma, and an RBF kernel, which represents the correlation function. The strength of each constituent kernel is then treated as a hyperparameter and subsequently tuned to provide the best performance.

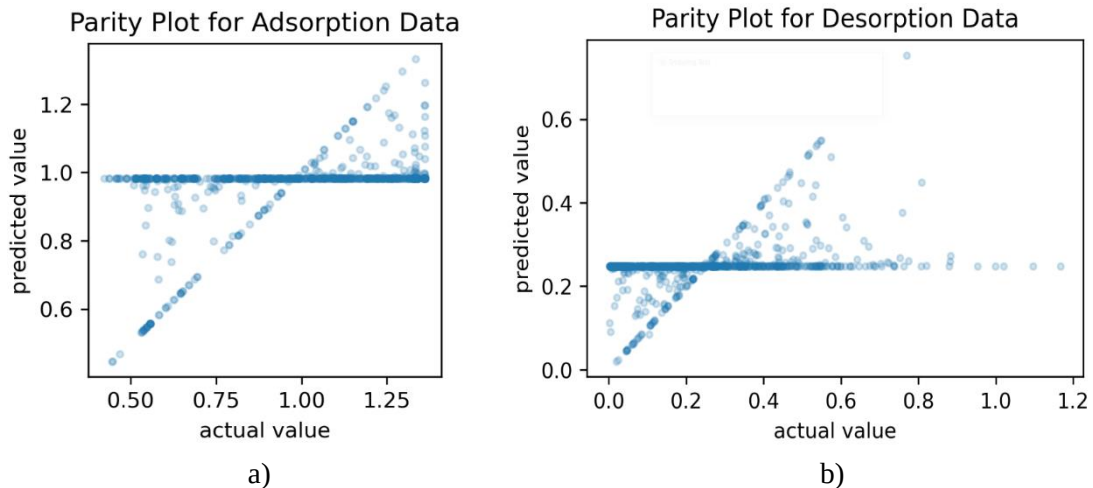


Figure 4. Parity plots on the validation set for the Kriging model of a) the quantity of adsorption output and b) the quantity of desorption output.

4.2 Automated Learning of Algebraic Models (ALAMO)

The ALAMO models were trained using ALAMOPY (ALAMO Python wrapper), which is a wrapper for the ALAMO software in Python. We chose to train polynomial models with ALAMO. Specifically, the surrogate models included monomial terms up to the third order and bivariate terms (squared or not), in addition to the constant and linear terms. The training times and R^2 values for the two outputs are presented in Table 3. It is observed that the training times are one order of magnitude less than the times required to train more complex surrogates. Additionally, the R^2 value for the quantity of desorption output is very good, standing at 0.97. This result along with the parity plot for the specific output in Figure 5b indicate that the quantity of desorption response surface is approximated well by a third-order polynomial model. However, for high values of this output variable, a noticeable deviation between the actual and predicted values is observed. As for the ALAMO surrogate for the quantity of adsorption output, its R^2 value stands at 0.88 and its parity plot is presented in Figure 5a. The polynomial surrogate approximates adequately well the output's response surface, but with notable deviations between certain actual and predicted values.

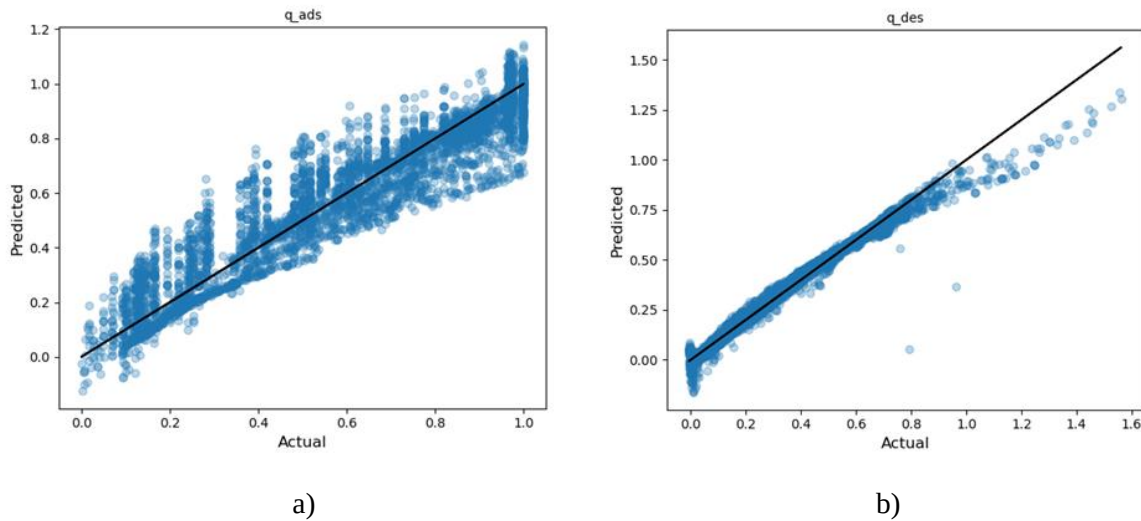


Figure 5. Parity plots on the validation set for the ALAMO models of a) the quantity of adsorption output and b) the quantity of desorption output.

The ALAMO polynomial surrogate models for the two outputs are given below:

$$\begin{aligned}
 q_{ads} = & 2.3 * T + 3.8 * Humidity + 0.43 * AdsTime - 0.76 * DesTime - 1.4 * SteamVelocity \\
 & - 1.2 * T^2 - 3.8 * Humidity^2 + 0.16 * AdsTime^2 + 0.98 * DesTime^2 + 1.9 * \\
 & SteamVelocity^2 + 0.22E-01 * T^3 + 1.5 * Humidity^3 - 0.46 * AdsTime^3 - 0.61 * DesTime^3 \\
 & - 0.95 * SteamVelocity^3 - 2.6 * T * Humidity + 0.38 * T * AdsTime - 0.29 * T * DesTime - 0.12 \\
 & * T * SteamVelocity - 0.90E-01 * Humidity * AdsTime - 0.24 * Humidity * DesTime - 0.31 * \\
 & Humidity * SteamVelocity + 0.65 * AdsTime * DesTime + 0.96 * AdsTime * SteamVelocity - 0.10 \\
 & * DesTime * SteamVelocity + 1.3 * (T * Humidity)^2 - 0.21 * (T * AdsTime)^2 + 0.12 *
 \end{aligned}$$

$$(T*DesTime)^2 - 0.40E-01 * (T*SteamVelocity)^2 + 0.18 * (Humidity*AdsTime)^2 + 0.23 * (Humidity*DesTime)^2 + 0.18 * (Humidity*SteamVelocity)^2 - 0.17 * (AdsTime*DesTime)^2 - 0.43 * (AdsTime*SteamVelocity)^2 - 0.61$$

$$q_{des} = 0.24E-01 * T - 0.17E-01 * Humidity + 0.12E-01 * AdsTime - 3.2 * DesTime - 2.4 * SteamVelocity - 0.82E-01 * T^2 + 0.13E-01 * Humidity^2 + 0.51E-01 * AdsTime^2 + 4.2 * DesTime^2 + 2.8 * SteamVelocity^2 + 0.56E-01 * T^3 - 0.49E-01 * AdsTime^3 - 2.1 * DesTime^3 - 1.3 * SteamVelocity^3 - 0.33E-01 * T*AdsTime + 0.19E-01 * T*DesTime - 0.92E-01 * AdsTime*DesTime + 0.21E-01 * AdsTime*SteamVelocity + 0.25 * DesTime*SteamVelocity + 0.12 * (AdsTime*DesTime)^2 + 0.25 * (DesTime*SteamVelocity)^2 + 1.5$$

4.3 Artificial Neural Network (ANN)

The ANN model was constructed with two hidden layers using sigmoid activation functions, which are well-suited for non-linear datasets (see equation below), featuring 64 and 32 neurons to introduce non-linearity. The final layer, with a single neuron and a linear activation function, was used for regression to predict continuous numeric values.

$$\sigma(x) = \frac{1}{(1+e^{-x})}, \text{ where } x = \text{input variables}$$

The model was trained using the Adam optimizer and mean squared error as the loss function, with 20% of the training data reserved for validation. A total of 100 epochs were used when fitting the model, yielding R² values of 0.886 and 0.981, which are comparable to those from ALAMO. Compared to ALAMO, ANN produced a worse parity plot for both output variables and required 5 to 6 times longer training time resulting in a lower aggregate result than ALAMO.

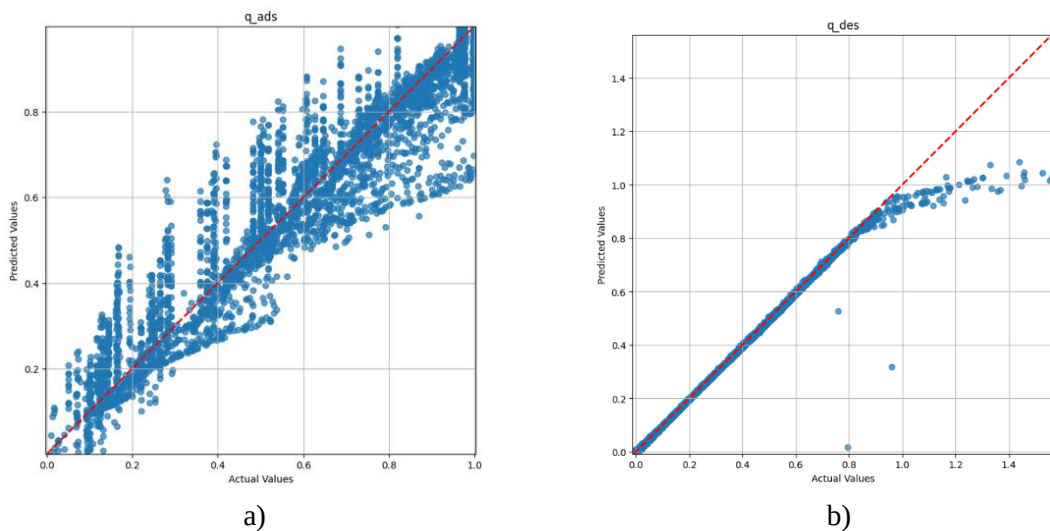


Figure 6. Parity plots on the validation set for the ANN models of a) the quantity of adsorption output and b) the quantity of desorption output.

4.4 Linear Regression using RBF kernel

The implementation of this method sought to represent the non-linear relationships by performing RBF feature transformation. Because of the high computational cost, only 20% of the winsorized training set was sampled and used for the LinearRegression fitting. Then, the rest 80% of the training set was used for optimizing the hyperparameter γ . Next, we used the best model to predict the validation set and evaluated R^2 and the parity plot. Each model yielded R^2 values of 0.954 and 0.964, respectively. While these models showed reasonable training time and R^2 , this surrogate model is not that ideal, as using a limited training set can increase the risk of overfitting and reduce the generalizability.

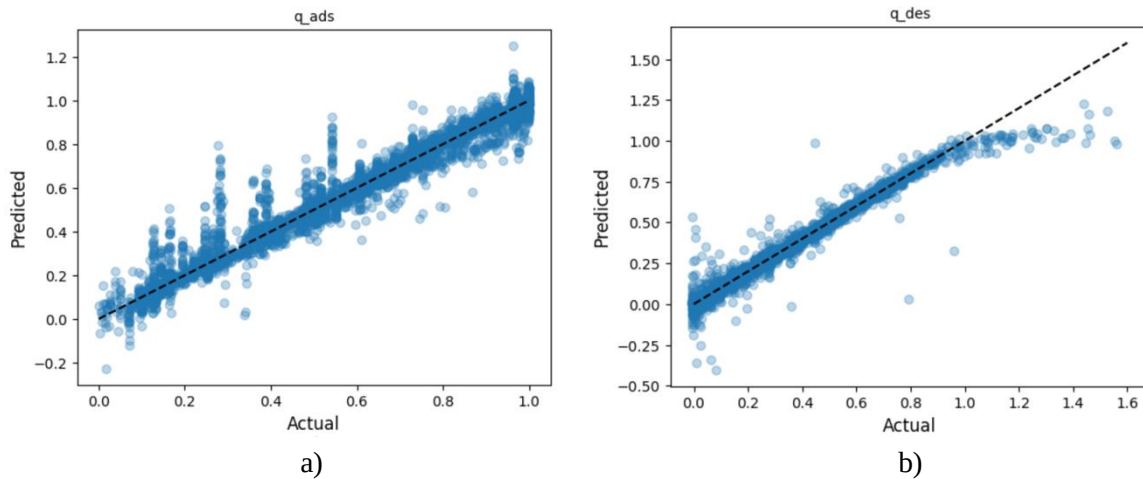


Figure 7. Parity plots on the validation set for the Linear Regression model using RBF transformation of a) the quantity of adsorption output and b) the quantity of desorption output.

4.5 Improved Model and Baseline Model Comparison

Our results indicate that kriging is not an appropriate method to approximate the sorption process, since the R^2 values and the aggregate score took considerably low values. The ALAMO, linear regression with RBF kernel and ANN techniques performed significantly better in terms of predictive ability in comparison to all the techniques studied. The ALAMO polynomial model achieved the highest aggregate score for the case of the quantity of desorption output and the RBF model was the best for the quantity of adsorption output, which indicates that these models balance predictive accuracy and implementation efficiency.

The ALAMO polynomial models have much improved performance metrics and parity plots in comparison to the baseline LASSO model. Specifically, for the quantity of adsorption output the R^2 values on the validation set for the LASSO and ALAMO surrogates stand at 0.78 and 0.88 respectively. As for the quantity of desorption output the R^2 values for the LASSO and ALAMO models are 0.71 and 0.97 respectively. The linear LASSO models require less training times and are simpler than the ALAMO polynomials as expected, but still the latter are computationally efficient, and their analytical expressions are not very complex, since they are cubic polynomials.

To further clarify the results from our improved ALAMO model and the baseline LASSO model, we performed a correlation analysis on the Winsorized training dataset, considering all input and output variables, as shown in the correlation matrix below.

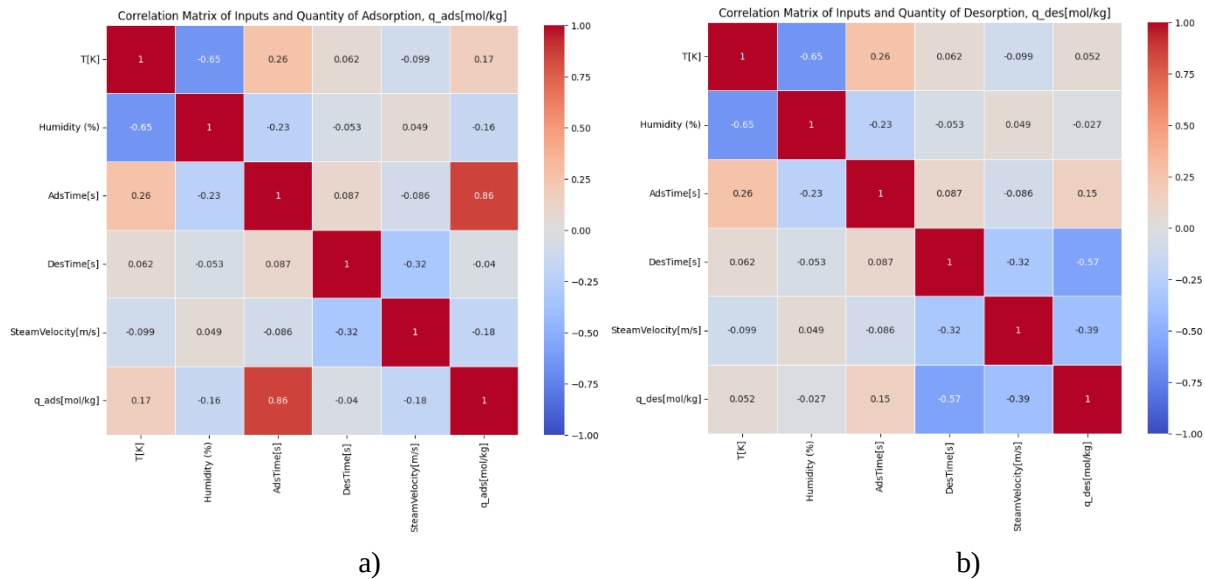


Figure 8. Correlation matrix of the input variables and a) the quantity of adsorption output and b) the quantity of desorption output.

As seen in Figure 8a), as expected adsorption time (AdsTime [s]) have the strongest correlation to quantity of adsorption (q_{ads} [mol/kg]). The same can be seen with desorption time (DesTime [s]) having the strongest correlation to quantity of adsorption (q_{ads} [mol/kg]), followed by steam velocity (SteamVelocity [m/s]). This pattern is consistent across the outputs from both the ALAMO and LASSO models, thereby solidifying our understanding of our dataset.

5. Conclusions

Given the size of our dataset, we knew going in that computation time was going to be a major factor in our model selection process; the aggregate score we developed did an acceptable job of quantifying model performance in a way that aligned with our expectations.

The ALAMO polynomial models significantly outperformed our baseline LASSO model for both adsorption (R^2 score of 0.88 vs 0.78) and desorption (0.97 vs 0.71). While linear LASSO models required less training time and were less complex, the ALAMO model remained computationally efficient up to 3rd-order polynomials. In contrast, Kriging proved both computationally inefficient and ineffective in practice. Although the Artificial Neural Network (ANN) showed promise with its R^2 score, it ultimately exhibited worse performance when visualized with a parity plot, which cannot be accounted for with our aggregate scoring method. Moreover, the ANN introduced unnecessary complexity, which is exhibited as significantly increased training time and requiring a higher degree of comprehension to understand the

premise of the model. RBF model also showed high aggregate scores, and the one for the adsorption model was the highest. However, the inherent risk of overfitting makes it difficult to claim that this model is optimal.

Finally, the correlation matrix analysis agrees with our results from the LASSO and ALAMO models, and it is shown that the adsorption time and desorption time inputs have the strongest correlation to quantity of adsorption and quantity of desorption outputs respectively.

5.1 Future Directions

A suggested future direction for our project involves embedding the surrogate model into an optimization framework. This would enable a comparison of the surrogate modeling techniques, specifically evaluating the trade-offs between predictive accuracy, complexity, and computational efficiency, and further highlight the effectiveness of polynomial surrogate models. For instance, by maximizing adsorption quantity per unit of energy, this approach could streamline decision-making in carbon capture processes.

Statement of Contributions

	Dimitrios Fardis	Hui Min Tee	Kyle Hoppner	Taewoo Kim	Joseph Ciambriello
Report Writing	X	X	X	X	X
Code Refactoring / Consolidation	X	X	X	X	X
Data Pre-processing		X	X		
Baseline Model	X				
ALAMO	X				
Kriging			X		
ANN		X			
RBF				X	X

References

Lee, W. H.; Zhang, X.; Banerjee, S.; Jones, C. W.; Realff, M. J.; Lively, R. P. Sorbent-Coated Carbon Fibers for Direct Air Capture Using Electrically Driven Temperature Swing Adsorption. *Joule* 2023, 7 (6), 1241–1259. <https://doi.org/10.1016/j.joule.2023.05.016>